L E N D I N G R I S K A R C H I T E C T U R E

Loan Origination & Management System

Comprehensive Schema

KYC / Onboarding · Fraud & Document Verification · Credit Scoring & Limits
Risk-Based Pricing · IFRS 9 Provisioning · Portfolio Monitoring · Collections & EWS
Plus system architecture for engineering and operations (services, events, data plane, security)

Version 1.0 | September 2026 | Reference architecture for design, data model, and metric catalog

Not a regulatory opinion. Adapt PD/LGD/EAD definitions, default definitions, and SICR policy to local supervisor and product book.

1. Purpose and design principles

This schema describes an end-to-end credit system in which no loan is originated, priced, booked, or collected without a measurable risk object attached to it. Each module produces scores, gates, and metrics that the next module consumes. The unit of work is the Application through to the Account and, after default, the Collection Case.

Five design principles:

- Gate before score. Identity, fraud, and document integrity must pass (or be explicitly overridden with audit) before credit models consume the file.
- One customer, many scores. Application score, bureau score, affordability score, fraud score, IFRS 9 PD, early-warning score, and collection propensity are separate objects with versioned models.
- Point-in-time and through-the-cycle live together. Origination and IFRS 9 use PIT, forward-looking parameters. Capital and pricing floors may use TTC overlays.
- Explainability is a first-class field. Every automated decision stores reason codes, feature contributions, policy rule hits, and model version.
- Monitoring is continuous, not a report. Portfolio and EWS scores refresh on a defined cadence (daily for performing book; intra-day for high-risk and collections).

2. End-to-end process and decision gates

2.1 Lifecycle

Lead / Pre-qualify → Application capture → KYC & identity → Document ingest & forensics → Fraud fusion → Affordability & bureau → Application credit score & limit → Risk-based price & product

offer → Underwriter / STP decision → Contract & disbursement → Booking (Stage 1 ECL) → Performing monitoring & EWS → SICR / Stage migration → Delinquency → Collection propensity & strategy → Cure, restructure, write-off, or recover.

2.2 Hard gates (must resolve before next stage)

Gate	Pass condition	Fail / exception path
G1 Identity	KYC status = VERIFIED; sanctions/PEP/adverse media clear or approved	Reject or enhanced due diligence queue
G2 Document integrity	Required pack complete; forensic score below reject threshold	Re-request docs or fraud investigation
G3 Fraud fusion	Composite fraud score < cutoff; no confirmed synthetic / first-party fraud	Decline or SIU referral
G4 Policy knockout	No hard policy fails (age, jurisdiction, prohibited sector, existing NPL, etc.)	Auto-decline
G5 Affordability	DSR / FOIR / DSCR within policy after verified income	Reduce limit or decline
G6 Credit decision	Application score + overrides produce APPROVE / CONDITIONAL	Decline; retain score for decline analysis
G7 Price & limit	Priced offer within risk-appetite bands; limit ≤ capacity	Re-price, downsell, or decline
G8 Disburse	Mandate, collateral, insurance, conditions precedent complete	Hold booking
G9 Stage 1 book	IFRS 9 PD/LGD/EAD assigned; 12-month ECL posted	Do not book until ECL engine returns

3. Logical data model (core entities)

Treat these as canonical objects. Physical tables can split history, scores, and documents. Every score row is append-only with model_version, as_of_ts, and feature_snapshot_id.

3.1 Party and identity

Entity	Key fields	Notes
PARTY	party_id, party_type (IND/ORG), status, created_ts	Golden customer record
PARTY_IDENTITY	id_type, id_number_hash, issuing_country, expiry, verify_status, source	Never store raw national ID in clear in analytics stores
PARTY_BIOMETRIC	face_match_score, liveness_score, capture_ts, vendor	vKYC evidence
PARTY_ADDRESS / CONTACT	type, value, verified_flag, geo, source	Velocity and device linkage use these
RELATED_PARTY	from_id, to_id, relation_type, ownership_pct	UBO, employer, guarantor, household

Entity	Key fields	Notes
SANCTIONS_SCREEN	list, match_score, disposition, analyst_id, as_of	PEP, AML, adverse media

3.2 Application, documents, fraud

Entity	Key fields	Notes
APPLICATION	app_id, party_id, product_id, channel, amount_requested, tenure, purpose, status	Status machine is explicit
APP_DECLARATION	income, employment, expenses, dependents, existing_debt	Stated vs verified stored separately
DOCUMENT	doc_id, app_id, doc_type, hash, mime, source (upload/AA/DigiLocker)	Immutable blob + metadata
DOC_FORENSIC	tamper_score, metadata_anomaly, template_match, vendor_score	One row per check run
FRAUD_SIGNAL	signal_code, severity, value, source	Device, velocity, graph, synthetic
FRAUD_SCORE	composite, first_party, third_party, synthetic, decision	Fusion layer output

3.3 Credit decision, account, collateral

Entity	Key fields	Notes
BUREAU_PULL	bureau, pull_type (soft/hard), score, tradelines_json, enquiries	Retain raw + parsed
CREDIT_SCORE	score_type, raw_score, pd_12m, grade, reason_codes[], model_id	APP, BEHAVIOURAL, BUREAU
AFFORDABILITY	verified_income, commitments, DSR, FOIR, residual, method	Stated vs verified dual track
LIMIT_DECISION	approved_limit, tenor, product, conditions[], decision	May downsell
PRICING	risk_band, base_rate, risk_premium, all-in rate, fees, APR/EIR	Risk-based + commercial caps
ACCOUNT	account_id, app_id, party_id, booked_ts, principal, EIR, status	Servicing system of record
SCHEDULE / PAYMENT	due_dt, due_amt, paid_amt, dpd, channel	Source of DPD and bounce
COLLATERAL	type, valuation, ltv, haircut, last_reval_dt, enforceability	Feeds LGD
GUARANTEE	guarantor_id, type, amount, rating	CRM / LGD credit

3.4 IFRS 9, monitoring, collections

Entity	Key fields	Notes
IFRS9_POSITION	account_id, as_of, stage, sicr_flag, default_flag	Daily or month-end snapshot
ECL_RUN	run_id, scenario, weight, pd, lgd, ead, ecl, discounted_ecl	Multi-scenario weighted
MACRO_SCENARIO	name, weight, horizon, variable set	Baseline / upside / downside
EWS_SCORE	score, band (G/A/R), drivers[],	Pre-delinquency

Entity	Key fields	Notes
	recommended_action	
COLLECTION_CASE	case_id, account_id, entry_dpd, strategy, owner	Opens at policy DPD or EWS red
COLLECT_SCORE	horizon, p_roll, p_cure, p_pay_today, expected_recovery	Pre and post variants
TREATMENT	channel, offer_type, cost, outcome, promise_dt	For strategy optimization
MODEL_REGISTRY	model_id, type, version, train_dt, champion/challenger, auc	Governance

4. Module A — KYC, onboarding, fraud, document verification

4.1 Objectives

Prove the person or entity is who they claim to be, is allowed to be served, submitted authentic evidence, and is not a synthetic, mule, or first-party fraud pattern. Output is a KYC case status plus a fraud composite that either blocks or annotates the credit file.

4.2 Process blocks

- Identity: government ID validation, biometric face match, liveness, video KYC where required, UBO and corporate registry for SMEs.
- Screening: sanctions, PEP, adverse media, internal watchlist, related-party graph.
- Document pack: required vs optional by product; completeness; expiry; source-of-truth preference (account aggregator / employer API over uploaded PDF).
- Forensics: file metadata, edit traces, template mismatch, reused images, inconsistent fonts/dates, EXIF vs claimed origin.
- Fraud fusion: device fingerprint, IP/geo vs KYC address, application velocity, graph (shared device/phone/employer), synthetic identity features, income/document inconsistency.
- AML/KYC refresh: periodic and event-driven re-KYC after booking (not only at origination).

4.3 Metric catalog — KYC / onboarding

Metric	Definition	Why it matters
KYC completion rate	Verified KYC / applications started	Funnel leakage vs control tightness
STP KYC rate	Auto-verified with no analyst / total KYC	Cost and cycle time
KYC cycle time (p50/p95)	Minutes from start to VERIFIED or FAIL	Drop-off driver
ID match rate	Successful bureau/gov ID match	Data quality + fraud
Face-match score distribution	Histogram of biometric scores	Threshold calibration
Liveness fail rate	Failed liveness / vKYC attempts	Presentation-attack pressure
EDD rate	Enhanced due diligence / KYC cases	Risk mix of book
Sanctions true-match rate	Confirmed hits / alerts	Alert quality
False positive screen rate	Cleared alerts / alerts	Ops load

Metric	Definition	Why it matters
Re-KYC overdue %	Parties past refresh SLA	Compliance residual risk
Beneficial owner completeness	Orgs with UBO chain complete	AML quality
Onboarding abandonment by step	Drop-off at ID, docs, funding	UX vs friction trade-off

4.4 Metric catalog — document verification

Metric	Definition	Why it matters
Pack completeness rate	Apps with all required docs accepted	Prevents underwriting on partial files
First-time-right doc rate	Accepted on first upload / uploads	Customer friction
Forensic flag rate	Docs with tamper/anomaly above warn	Attack surface
Confirmed doc fraud rate	SIU-confirmed / flagged	Model precision
Source mix	% AA / employer API / upload / DigiLocker	Verification quality of book
OCR field error rate	Manual corrections / extracted fields	Data quality into score
Statement months coverage	Avg months of bank data used	Income model robustness
Doc-to-declaration mismatch rate	Income/ID fields disagree with stated	First-party fraud + error

4.5 Metric catalog — fraud

Metric	Definition	Why it matters
Composite fraud score	Fused 0–1000 or PD-like	Gate G3
Synthetic ID score	Inconsistency of ID, credit file thinness, date collisions	High-loss fraud type
Velocity: apps per device/phone/ID 24h/7d/30d	Count + unique products	Burst attacks
Graph cluster risk	Shared attributes with known fraud	Network fraud
Device reputation / emulator flag	Vendor + internal	Bot and farm
First-party fraud rate (vintage)	Confirmed misrep / booked loans by MOB	Origination quality
Third-party fraud rate	Identity takeover / booked	KYC hole
Fraud loss / disbursement	Net fraud $ / $ booked	Economic control
Fraud model precision@cutoff	Confirmed / auto-declined as fraud	Avoid good-customer decline
Override rate on fraud decline	Analyst force-through / fraud declines	Governance
Time-to-detect fraud (post-book)	Days from book to SIU confirm	Monitoring lag

4.6 Suggested feature groups for fraud model

- Identity consistency: name/DOB/ID/phone/email/address history coherence.
- Digital: device ID, rooted/jailbreak, emulator, IP proxy/VPN, geo vs KYC, session duration.
- Application behaviour: typing cadence, copy-paste of IDs, midnight applications, repeated retries.
- Financial document: round-number salaries, identical PDF templates across unrelated apps, edited bank lines.

- Credit file: thin file + high limit request, burst of new tradelines, enquiry spike.
- Network: same employer string + same device across many new-to-bank apps.

5. Module B — Credit scoring, limits, risk-based pricing

5.1 Score objects (keep them separate)

Score	Predicts	Used for
Application / origination PD	P(default within definition window, e.g. 90+ DPD in 12 months)	Approve/decline, grade, limit, price
Bureau score (external)	Generic credit risk	Knockout, overlay, dual-score policy
Affordability / capacity	P(payment stress) and max instalment	Limit cap, tenor
Limit model	Optimal exposure given PD, LGD, capital, margin	Approved amount
Behavioural score (post-book)	Forward PD given performance	Limit management, EWS, staging overlay
Response / take-up score	P(accept offer)	Price elasticity, offer design
Fraud-adjusted PD	Credit PD conditioned on residual fraud risk	Conservative book where fraud uncertain

5.2 Application scorecard — typical feature families

- Bureau: score, worst status, utilisation, enquiries 3/6/12m, age of file, unsecured balances, recent delinquencies.
- Affordability: verified income, residual income, DSR/FOIR, expense ratio, dependents, employment tenure, sector/employer stability.
- Banking behaviour (AA/statements): salary regularity, buffer vs EMI, NSF/bounce count, gambling/cash intensity, inflow volatility.
- Application: requested amount vs income, purpose, channel, existing relationship tenure, product mix.
- Collateral (secured): LTV, asset liquidity, valuation confidence, insurance.
- SME extras: vintage of firm, GST consistency, bank credits vs declared turnover, director bureau, concentration of buyers.
- Alternative (where lawful): utility/telco, psychometric only with strong governance, mobile money float — never as sole gate).

5.3 Limit engine

Approved limit is the minimum of policy max, affordability max, collateral max, concentration caps, and risk-appetite max implied by PD and target loss.

```
Limit = min( PolicyMax, AffordabilityMax, CollateralMax(LTV),
ConcentrationHeadroom, RiskCap(PD, LGD, TargetEL, Capital) )
```

RiskCap example (illustrative, not a pricing theorem): size the loan so expected loss and capital load still clear hurdle ROE after funding and opex. Store every binding constraint as a reason code (AFFORDABILITY_BIND, LTV_BIND, PD_BIND).

5.4 Risk-based pricing

Price is a function of funding cost, operating cost, expected loss, capital cost, and strategic margin, subject to regulatory caps and fairness constraints.

```
All-in rate ≈ Funding + Opex + EL% + CostOfCapital% + Margin  ±  elasticity adjustment

EL% ≈ PD_horizon × LGD × EAD_factor   (aligned to product life, not necessarily IFRS 9 12m)
```

Implementation notes:

- Map application grade → risk band → price grid. Do not let the neural net emit a rate without a band and floor/ceiling.
- Separate credit price from commercial discount. Discounts require authority matrix and P&L attribution.
- Verification quality can be a pricing input: AA-verified income may price tighter than self-uploaded statements (document as a risk differential, not a channel tax).
- Monitor adverse selection: take-up rate by band vs subsequent PD. If only high-PD customers accept a price, the grid is wrong.
- Fair lending: disparate impact tests on protected attributes even if those attributes are not model inputs (proxies via geography/channel).

5.5 Metric catalog — credit decisioning

Metric	Definition	Why it matters
Approval rate	Approves / decisioned apps	Appetite vs volume
STP decision rate	No human credit touch / decisioned	Efficiency
Override rate (approve↔decline)	Human ≠model / decisioned	Policy leakage
Downsell rate	Offered < requested / approves	Limit engine bind
Take-up rate	Accepted booked / approved offers	Price & process
Time-to-yes p50/p95	App start to decision	Conversion
Average approved PD / EL	Exposure-weighted	Book quality at gate
KS / Gini / AUC (out-of-time)	Rank-order of defaults	Model power
PSI / CSI	Population & characteristic shift	Drift
Grade concentration	% in each risk grade	Risk appetite
First-payment default / MOB3 DQ	Early default by vintage	Origination defect
Through-the-door vs booked PD gap	Selection effect	Adverse selection
Limit utilisation at 90 days	Drawn / approved (revolving)	EAD insight

Metric	Definition	Why it matters
Price vs band slippage	Actual APR − grid APR	Discount leakage
Hurdle ROE hit rate	Deals clearing target ROE	Pricing discipline

6. Module C — IFRS 9 staging and loan provisioning

6.1 Staging (general approach)

Stage	Credit condition	Loss allowance
Stage 1	Performing; no SICR since origination	12-month ECL
Stage 2	SICR since origination; not credit-impaired	Lifetime ECL
Stage 3	Credit-impaired / defaulted (internal default definition)	Lifetime ECL; interest on net carrying amount

Default definition must be documented (typical: 90 DPD, unlikeliness-to-pay, specific provisions, distressed restructure). Cure and probation rules must be explicit so accounts do not oscillate stages.

6.2 Core formula

```
ECL = Σ_t  [ mPD_t  ×  LGD_t  ×  EAD_t  ×  Survival_{t-1}  /  (1 + EIR)^t ]
```

Reported ECL is the probability-weighted average across macroeconomic scenarios (baseline, upside, downside). Stage 1 uses default events in the next 12 months; Stages 2 and 3 use the remaining contractual life, including expected prepayments and, for revolvers, expected drawdown via CCF.

```
Revolver EAD ≈ Drawn + CCF × Undrawn   (CCF from defaulted-path history, PIT)
```

6.3 SICR framework (store every trigger)

- Relative PD test: lifetime PD now vs lifetime PD at origination exceeds a multiple and/or absolute gap (policy numbers belong in a governed parameter table).
- Absolute backstops: 30 DPD (IFRS 9 rebuttable presumption), watchlist, forbearance, rank downgrade of X notches.
- Low-credit-risk expedient: only if policy allows and instrument meets investment-grade equivalent.
- Collective vs individual assessment: small retail on scorecards; large exposures on expert + model.
- Forward-looking overlay: sector, geography, employer shock not yet in DPD.

6.4 Parameter objects

Parameter	Origination use	IFRS 9 use
PD	TTC or hybrid for pricing floors; PIT for decision	PIT, forward-looking, term structure

Parameter	Origination use	IFRS 9 use
LGD	Downturn-leaning for price/capital	Best-estimate PIT; recoveries net of costs; collateral paths
EAD / CCF	Limit + expected draw	Path to default; prepayments; undrawn
EIR	Product APR build-up	Discount rate for ECL cash shortfalls
Scenarios	Stress for appetite	Weighted ECL; IFRS 7 sensitivity

6.5 Metric catalog — provisioning & credit quality

Metric	Definition	Why it matters
Stage mix (count & $)	% Stage 1/2/3 of book	Headline credit quality
SICR rate / new Stage 2 $	Inflows to Stage 2 in period	Emerging stress
Cure rate Stage 2→1, 3→2	With probation filter	Forbearance quality
Coverage ratio by stage	ECL / gross exposure	Adequacy
ECL / GCA (overall)	Stock of provisions	P&L and capital
ECL charge (P&L)	Δ stock − write-offs + recoveries (sign per policy)	Earnings volatility
Coverage vs NPL	Stage 3 ECL / Stage 3 exposure	Loss given default realised vs modelled
PD backtest (12m)	Predicted vs realised default by grade	Model validity
LGD backtest	Predicted vs closed-recovery	Recovery process + model
EAD / CCF backtest	Predicted vs drawn-at-default	Revolver risk
Overlay stock	Management / post-model adj $	Judgement vs model
Scenario weight sensitivity	ECL if downside weight +10pp	Disclosure & ICAAP
Write-off rate	Write-offs / average book	Loss realisation
NPL / PAR30 / PAR90	Standard portfolio DQ	Ops + regulator language

7. Module D — Loan and portfolio monitoring

7.1 Two clocks

Account clock: this borrower, this facility — payments, cash-flow, bureau refresh, collateral, covenant, contact. Portfolio clock: vintage, product, channel, grade, geography, officer, employer/sector — concentration and quality drift of the factory that originated the book.

7.2 Account-level monitoring objects

- Repayment: DPD, days-early, partial pays, bounce/NACH fail count, payment channel mix.
- Behavioural score refresh (monthly or event-driven).
- Open-banking / AA refresh where consented: salary delay, buffer/EMI, essential vs discretionary spend, new loan credits from other lenders.
- Bureau refresh: new enquiries, new tradelines, worsening status elsewhere (especially useful before first missed EMI).

- Collateral: valuation lag, insurance lapse, LTV breach.
- SME: GST filings missed, covenant DSCR, auditor flags, related-party deposits.
- Operational: returned mail, phone disconnected, app login drop, complaint surge.

7.3 Portfolio metric catalog

Metric	Definition	Cut by
Book $ and count	Gross / net carrying amount	Product, grade, stage, channel
Disbursement volume	New $ in period	Vintage month
PAR30 / PAR90	Balance ≥30/90 DPD / book	Vintage, MOB, officer
Flow rates 0–30–60–90–WO	Roll-rate matrix	Product, grade
Vintage DQ curve	DQ% by months-on-book	Origination month, score band
CDR / default rate	Defaulted principal / start principal	Segment
CPR / prepay rate	Conditional prepayment	Rate environment, product
Net credit loss	Write-off − recovery	Segment
Yield / NIM / risk-adjusted margin	Interest − EL − funding	Band
Concentration HHI	Sector, geography, employer, single name	Book
Limit breach count	Policy / covenant / LTV	Portfolio
Model PSI by vintage	Applicant mix shift	Channel
Officer / branch PAR gap	Unit PAR / book PAR	Agency risk
Restructure stock %	Modified loans / book	Hidden stress

7.4 Vintage and MOB are mandatory views

A blended PAR can hide a bad new vintage under an old good book. Every credit committee pack should show MOB 1/3/6/12 default or 30 DPD by origination month, score band, channel, and verification type (AA vs upload). First-payment-default and MOB3 are origination defect indicators; MOB12+ is more about economy and servicing.

8. Module E — Early warning and collection scores

8.1 Split the prediction problems

Do not use one score for everything. Pre-disbursement collection risk is an origination overlay. Post-disbursement EWS is a performing-book warning. In-collections scores are operational allocation tools. They share features but have different labels, horizons, and actions.

Score	Population	Label (example)	Primary action
Pre-disburse collection propensity	Applicants passing credit gates	P(ever 30+ or 90+ in first 12 MOB); P(recover | default)	Price, limit cut, extra collateral, decline gray zone
EWS / delinquency propensity	Performing accounts (DPD 0–x)	P(miss next EMI); P(30+ in 30–90 days)	RM contact, restructuring offer, limit freeze

Score	Population	Label (example)	Primary action
Roll-rate / severity	Accounts already at DPD x	P(reach x+y | at x)	Strategy intensity
Cure propensity	Delinquent not charged-off	P(return to current in 30/60d)	Self-cure wait vs work
Pay-today / contact propensity	Assigned collection queue	P(payment or kept PTP | contact today)	Queue ranking
Recovery / LGD-in-work	Defaulted / Stage 3	Expected $ recovered by horizon, channel	Settle vs legal vs sale
Skip / contactability	All worked accounts	P(valid phone/address)	Data enrichment spend

8.2 Pre-disbursement collection score

This is not a second credit PD. Credit PD answers “will they default?”. Pre-disburse collection score answers “if this file later needs collections, how ugly will it be, and how collectible is this person?” Thin-file applicants with high PD but salary-domiciled accounts and reachable contact graph may still be operationally collectible; cash-heavy, multi-SIM, no-trace profiles are not.

Inputs beyond credit PD: contact quality (phone/email/employer verified), stability (address tenure, job tenure), bank domicile with lender, existing collection history on other products, collateral enforceability, geography legal friction, household related-party performance, device persistence.

Outputs stored on APPLICATION and copied to ACCOUNT at book: p_ever_30_12m, p_ever_90_12m, p_right_party_contact, expected_recovery_rate_if_default, recommended_collateral_addon, recommended_mandate_type (auto-debit vs cash).

8.3 Post-disbursement EWS (before DPD)

Refresh daily. Band Green / Amber / Red with mandatory actions and SLAs. A Red EWS account is not yet a collection case; it is a prevention case.

Leading signals that typically move 30–45 days before first miss:

- Salary credit late or missing; buffer balance falling below 1× EMI.
- Failed or cancelled auto-debit registration; repeated low-balance before due date.
- New unsecured enquiries or new loans elsewhere.
- Utilisation spike on revolving products; minimum-pay behaviour.
- App engagement drop; returned communications; phone invalid.
- Employer mass-layoff or sector shock list; GST stop-filings for SMEs.
- Promise patterns on other products; hardship / complaint tags.

8.4 In-collections scores

Once policy DPD (or Red EWS + miss) opens a case, rank work by value of action, not by DPD alone. A DPD 45 with high pay-today score and recent salary credit outranks a silent DPD 20.

```
Work priority ≈ Expected cash from action − Cost of action − Customer franchise penalty
```

Treatments to log: SMS, USSD, app nudge, outbound call, field visit, restructure, settlement, legal, sale. Each treatment has cost, capacity, and empirically estimated lift. Champion-challenger the strategy tree.

8.5 Metric catalog — EWS and collections

Metric	Definition	Why it matters
EWS coverage	% performing book scored <24h stale	Operational readiness
Alert precision	% Amber/Red that miss EMI in 30–60d	Avoid cry-wolf
Alert recall	% of future 30 DPD that were Amber/Red prior	Early enough?
Lead time	Median days from first Red to first miss	Action window
Contact-within-SLA	First contact / alerts by band	Ops discipline
Pre-delinquency cure	Red that do not hit DPD1	Value of EWS
Roll rate x→x+30	By score decile	Score power in collections
Cure rate 30/60/90	Return to current	Strategy quality
PTP kept rate	Kept promises / promises	Agent + model
Right-party contact rate	RPC / attempts	Data + channel
Cost to collect / $ recovered	Direct ops cost	Efficiency
Recovery curve	Cumulative $ / default $ by months-since-default	LGD calibration
Legal vs soft recovery mix	$ and cycle time	Strategy
Re-age / restructure recidivism	Re-default in 6/12m	Forbearance abuse
NPA entry from watchlist	Watch → NPL trend	Board metric

9. Score fusion and the decision record

At origination the decision engine should persist a single CREDIT_DECISION_RECORD:

- Inputs: KYC status, fraud composite + top signals, doc integrity, policy knockouts, application PD + grade, affordability max, collection propensity, collateral LTV.
- Outputs: decision, limit, tenor, price band, actual price, conditions precedent, IFRS 9 origination PD/LGD used for day-1 ECL.
- Explain: reason codes (max 5), rule hits, model versions, override flag + approver + justification.
- Fairness snapshot: inferred segment tests stored at portfolio level, not as a borrower stigma field.

Post-book, a daily ACCOUNT_RISK_SNAPSHOT stores stage, ECL, behavioural PD, EWS band, collection scores if any, and top drivers. This snapshot is what monitoring, finance, and collections all read so they do not recompute competing truths.

10. Additional modules that improve the system

10.1 Model risk management

Model inventory, champion-challenger, approval committee, development/validation/monitoring pack (AUC, KS, PSI, calibration belt, stability of reason codes), challenger always running on 10–20% shadow. Outcome: no silent model swap.

10.2 Data quality and lineage

Field-level completeness, freshness, vendor SLA, reconciliation of servicing balances to ECL EAD. Metrics: % critical features null at decision, bureau pull fail rate, AA consent fail rate. Bad data is a credit risk.

10.3 Consent, privacy, and audit

Purpose-limited use of AA and bureau data; retention schedules; immutable decision log; maker-checker on overrides and stage overlays. Metrics: consent expiry, access log anomalies.

10.4 Collateral and valuation operations

AVM vs physical valuation variance, revaluation SLA, insurance lapse rate, enforcement cycle time. Direct LGD lever.

10.5 Limit management after booking

For revolvers: behavioural re-score, limit increase/decrease, freeze on EWS Red. Metrics: limit-increase subsequent default vs control.

10.6 Forbearance and hardship

Separate queue from ordinary collections. Track recidivism, Stage 2 overlays, and whether hardship is temporary (income shock) vs structural. Prevents silent evergreening.

10.7 Recoveries and NPL sale

Work-in-progress inventory aged by months-since-default; bid vs hold model using recovery curve vs sale price.

10.8 Capital, ICAAP, and stress

Reuse IFRS 9 engines with downturn LGD and stressed PD for ICAAP. Do not maintain a third disconnected PD universe without a documented bridge.

10.9 Conduct and complaints

Collections intensity caps, vulnerable-customer flags, complaint rate by strategy. Conduct failure becomes credit loss through remediation and franchise damage.

10.10 Channel and agency quality

Broker/officer early-vintage DQ, fraud rate, override rate. Clawback rules tied to MOB3/MOB6 defects.

11. Recommended operational cadences

Process	Cadence	Owner
KYC / fraud on new apps	Real time	Onboarding + SIU
Credit decision + price	Real time / near-real-time	Credit + pricing
Day-1 ECL on booking	At book event	Finance / risk ops
EWS + behavioural score	Daily	Credit risk / collections
Bureau / AA refresh (consented)	Monthly or event	Risk ops
IFRS 9 official run	Month-end + intra-month flash	Finance
SICR & watchlist committee	Weekly	Credit risk
Vintage & model monitoring pack	Monthly	MRM + credit
Collections strategy review	Weekly ops / monthly model	Collections
Model validation	At least annual + on drift breach	Independent validation

12. Minimal policy parameters to table-drive (do not hard-code)

- Default definition, DPD backstops (30/90), cure probation days.
- SICR relative and absolute PD thresholds by product.
- Fraud and forensic cutoffs (auto-decline / review / pass).
- Affordability caps (DSR/FOIR/DSCR) by product and segment.
- Price grid by grade × product × LTV bucket; min/max APR.
- EWS thresholds and contact SLAs by band.
- Collections strategy tree: which score decile gets which treatment at which DPD.
- Scenario weights and overlay governance.
- Concentration caps: single name, employer, sector, geography, channel.

13. Implementation sequence (practical)

If building rather than replacing everything at once:

- 1. Canonical PARTY, APPLICATION, ACCOUNT, PAYMENT, DOCUMENT — without these, scores have nowhere to live.

- 2. KYC + document completeness + simple rule fraud (velocity, watchlist) as hard gates.
- 3. Interpretable application scorecard + affordability + price grid. Log every decision.
- 4. Servicing events → DPD, roll rates, vintage dashboards.
- 5. IFRS 9 staging with pragmatic SICR (30 DPD + watchlist + PD multiple) and a documented PD/LGD/EAD v1.
- 6. EWS on internal payment + bounce + utilisation; then add AA/bureau refresh.
- 7. Collection propensity and queue ranking; treatment experiments.
- 8. Only then deep-learning fusion, graph fraud, and highly segmented LGD. Complexity without labels and ops uptake is waste.

14. Worked numeric sketch (illustrative only)

Book a 12-month unsecured loan of 100,000 at EIR 18%. Origination PIT 12-month PD 4.0%, LGD 65%, EAD 95,000 (allowing modest prepayment before default).

```
12-month ECL ≈ 95,000 × 0.04 × 0.65 = 2,470   (then discount at EIR to reporting date)
```

If six months later lifetime PD has doubled and 30 DPD is hit, SICR fires, account moves to Stage 2, remaining-life PD term structure is applied, LGD may worsen if collateral-less and cure probability drops, and ECL steps up materially. The same account's EWS should have been Amber/Red before the 30 DPD backstop if salary buffer and bounce signals were live. Collections then consume p_cure and p_pay_today rather than treating all 30 DPD as equal.

This sketch is pedagogical. Production ECL must use monthly marginal PDs, survival, prepayment, multiple scenarios, and the institution's approved default definition.

15. Glossary of score and risk terms

Term	Meaning in this schema
PD	Probability of default over a stated horizon or lifetime term structure
LGD	Loss given default after recoveries and costs, as % of EAD
EAD	Exposure at default, including expected draw of undrawn
CCF	Credit conversion factor on undrawn commitments
EIR	Effective interest rate used to discount IFRS 9 cash shortfalls
SICR	Significant increase in credit risk since initial recognition
PIT / TTC	Point-in-time vs through-the-cycle PD philosophy
EL	Expected loss, typically PD×LGD×EAD; pricing uses a product-life view
PAR30	Portfolio at risk: balances with DPD ≥ 30
MOB	Months on book since origination / first disbursement
PSI	Population stability index for model input mix

Term	Meaning in this schema
STP	Straight-through processing with no human credit touch
EWS	Early warning system on performing or early-delinquent book
PTP	Promise to pay in collections
FOIR / DSR	Fixed-obligation / debt-service ratio vs income

16. System architecture (for developers and system admins)

This section is the build-and-run view of the same schema. It is a reference architecture, not a mandated vendor stack. Swap Postgres for an existing core-banking ledger if the institution already has one; do not invent a second system of record for principal, interest, and DPD.

16.1 Architecture principles

- Bounded contexts over a monolith of everything, but a modular monolith is acceptable until team and scale force a split. Do not start with 18 microservices and no domain language.
- Write path is transactional and synchronous only where the customer is waiting (KYC, decision). Heavy work (OCR, bureau waterfall, ECL batch, EWS rescoring) is asynchronous behind an event bus.
- One system of record per fact. Servicing owns balances and DPD. Decision service owns the decision record. Document service owns blobs and forensic results. Scores are immutable append-only facts.
- Outbound vendor calls go through an integration adapter with timeouts, circuit breakers, idempotency keys, and a vendor-simulator for lower environments.
- PII and raw ID numbers never land in the warehouse, logs, or feature store in clear. Hash or tokenise at the edge.
- Every state change emits a domain event. Dashboards, IFRS 9, EWS, and collections subscribe; they do not scrape another service's database.

16.2 Logical layers

Layer	Responsibility	Typical components
Channel	Capture applications, vKYC, servicing self-serve, collector workspace	Mobile/web apps, agent portal, partner API, USSD
Edge	Authn/z, WAF, rate limit, mTLS, request ID	API gateway, IdP, bot/device fingerprint
Experience APIs	BFF per channel; orchestrate use-case calls	Origination BFF, Collections BFF
Domain services	Own entities and invariants listed in §3	See service catalog below
Decision & ML	Rules + models + explanations + registry	Decision engine, model serving, feature store
Integration	Vendors and core systems	Adapters: KYC, bureau, AA, payments, CBS, SMS
Async fabric	Events, jobs, outbox	Kafka/NATS + workers + scheduler
Data	OLTP, objects, search, analytics, features	See data plane
Control plane	Config, policy parameters, secrets, deploy	GitOps, parameter service, vault

16.3 Service catalog (bounded contexts)

Service	Owns	Sync API (examples)	Emits
party-identity	PARTY, identity, biometric refs, screening cases	POST /parties, POST /kyc/sessions, GET /parties/{id}/kyc	PartyCreated, KycVerified, KycFailed, ScreenHit
document-intel	DOCUMENT, forensic results, OCR fields	POST /docs, GET /docs/{id}, POST /docs/{id}/verify	DocumentReceived, ForensicCompleted, PackComplete
fraud	signals, graph edges, composite score	POST /fraud/score, GET /fraud/{appId}	FraudScored, FraudCaseOpened
origination	APPLICATION, declarations, status machine	POST /applications, PATCH status, GET case file	ApplicationSubmitted, ApplicationCancelled
decision	CREDIT_DECISION_RECORD, rules, overrides	POST /decide, POST /override	CreditDecided, OfferIssued, OverrideRecorded
pricing-limit	grids, limit calc, EIR/APR build-up	POST /price, POST /limit	PriceQuoted, LimitBound
agreement	e-sign, mandate, conditions precedent	POST /contracts, POST /mandates	ContractSigned, CpSatisfied
servicing (SoR)	ACCOUNT, schedule, payment, DPD	GET /accounts, POST /disburse (or CBS)	AccountBooked, PaymentPosted, PaymentBounced, DpdChanged
collateral	valuations, LTV, insurance	POST /collateral, POST /revalue	CollateralUpdated, LtvBreached
ifrs9-ecl	stage, ECL runs, scenarios	POST /ecl/run, GET /positions	StageMigrated, EclPosted
monitoring-ews	EWS scores, watchlist	GET /ews/{accountId}	EwsBandChanged, Watchlisted
collections	cases, treatments, PTP, scores	POST /cases, POST /treatments	CaseOpened, PtpRecorded, TreatmentApplied
model-platform	registry, features, scoring runtime	POST /score/{modelKey}	ModelScored, DriftAlerted
workflow	human queues, SLA, maker-checker	task APIs	TaskAssigned, TaskCompleted
notification	SMS, email, push, in-app	POST /notify	NotificationSent
audit-ledger	immutable decision and access log	append-only internal	AuditAppended

Servicing may be the existing core banking system (CBS). If so, origination books via a controlled adapter and consumes payment/DPD events from CBS. Never duplicate accrued interest in origination.

16.4 Reference runtime topology

Internet / partners → WAF + API gateway (JWT/mTLS, rate limits, correlation-id) → channel BFFs → domain services. Services talk to each other via sync HTTP/gRPC only for request/response on the customer path, and via the event bus for everything else. Workers sit off the bus: forensic worker, bureau waterfall worker, ECL engine, nightly EWS scorer, collection dialler feed, warehouse loader.

ASCII view of the happy origination path:

```
Channel → BFF → origination.create
  → party-identity.startKyc  → vendors (eKYC, liveness, sanctions)
  → document-intel.ingest     → object store + forensic worker
  → fraud.score               → graph + device + velocity
  → decision.decide           → bureau adapter + model-platform + pricing-limit
  → agreement + mandate       → disbursement adapter → CBS/servicing
  → events: AccountBooked     → ifrs9-ecl day-1 + warehouse + EWS seed
```

16.5 Data plane

Store	What lives there	Notes for admins
OLTP (PostgreSQL or CBS + Postgres sidecar)	Party, application, decision, case, parameters	Strong consistency; row-level security by tenant/branch if needed
Object store (S3/MinIO)	Raw docs, vKYC video, model artefacts	SSE-KMS; bucket per class; no public ACLs; signed URLs
Document metadata DB	Hashes, forensic JSON, retention class	Separate from blob; legal hold flag
Event log (Kafka)	Domain events, outbox relay	Compacted topics for current score; retention on PII topics short
Feature store (Feast/Tecton/Redis+PG)	Online features for scoring	Point-in-time correctness for training
Model registry + artefact store	Versions, cards, thresholds	Immutable; promote via GitOps
Search (OpenSearch)	Case file retrieval, collector search	Redact national IDs
Warehouse / lakehouse	Snapshots, vintages, ECL traces, features	No raw ID; hashed party_id
Secrets (Vault/cloud KMS)	Vendor API keys, DB creds, signing keys	Short TTL; no secrets in app config maps
Redis / cache	Sessions, idempotency keys, rate counters	Not a source of truth for balances

Outbox pattern (mandatory)

Domain service writes business row + outbox row in one DB transaction. A relay publisher pushes to Kafka. Consumers are idempotent on event_id. This avoids dual-write loss between Postgres and the bus.

Suggested core OLTP schemas (logical)

party, party_identity, party_address, related_party, sanctions_screen, application, application_status_hist, app_declaration, document, doc_forensic, fraud_signal, fraud_score,

bureau_pull, credit_score, affordability, credit_decision, pricing_quote, account, repayment_schedule, payment, dpd_snapshot, collateral, ifrs9_position, ecl_run, ecl_scenario_result, ews_score, collection_case, collection_score, treatment, model_registry, policy_parameter, audit_event, integration_outbox, idempotency_key.

Physical DDL should add: tenant_id if multi-lender, created_at/updated_at, created_by, row_version, and for scores: model_id, model_version, as_of_ts, feature_snapshot_id, reason_codes jsonb.

16.6 Event catalog (minimum)

Event	When	Key payload	Typical consumers
ApplicationSubmitted	Customer submits	app_id, party_id, product, amount	KYC, docs, fraud
KycVerified / KycFailed	Identity case closes	party_id, method, scores	origination, decision
ForensicCompleted	Doc check done	doc_id, tamper_score	origination gate
FraudScored	Fusion complete	app_id, composite, bands	decision, SIU workflow
BureauReceived	Vendor return	app_id, bureau, score	decision, feature store
CreditDecided	Approve/decline/refer	decision, limit, price, reasons	agreement, notify, warehouse
AccountBooked	Funds + account exist	account_id, principal, EIR	IFRS 9, EWS seed, warehouse
PaymentPosted / Bounced	Servicing	amount, due_dt, channel	DPD engine, EWS, collections
DpdChanged	DPD bucket moves	old, new, balance	IFRS 9 SICR, collections
StageMigrated	1↔2↔3	from, to, trigger	finance, RM, collections
EwsBandChanged	G/A/R change	band, drivers	notify, workflow, freeze limits
CaseOpened	Collections start	case_id, entry_dpd	dialler, collector UI
EclPosted	Run complete	run_id, stage, ecl	GL adapter, warehouse

Event envelope: event_id (ULID), event_type, occurred_at, producer, tenant_id, aggregate_type, aggregate_id, schema_version, correlation_id, causation_id, payload. Schema-registry the payload. PII fields encrypted or omitted; consumers that need PII call party-identity with authorisation.

16.7 Integration architecture

Adapter	Direction	Pattern	Ops concerns
eKYC / ID authority / DigiLocker	Outbound	Sync on customer path; cache evidence id	Legal retention; vendor SLA; sandbox identities
Liveness / face match	Outbound	Sync; store vendor reference not raw biometrics if policy allows	GPU vendor quotas; spoof updates
Sanctions / PEP / media	Outbound	Sync + nightly rescreen of book	Fuzzy-match alert volume
Credit bureaus	Outbound	Soft pull pre-qual; hard pull pre-decision; waterfall	Consent, enquiry footprint, cost
Account aggregator / open banking	Outbound	Consent grant then async data ready	Consent expiry jobs; refresh tokens

Adapter	Direction	Pattern	Ops concerns
Payroll / employer / GST	Outbound	Sync or batch	Employer coverage gaps
CBS / ledger / payments	Both	Book, disburse, collect, refund via idempotent API	Reconciliation every day; NACK handling
Mandate / NACH / debit order	Both	Register then listen to presentation results	Fail codes feed EWS
SMS / email / WhatsApp / dialler	Outbound	Async; preference + quiet hours	DND, conduct caps
GL / finance	Outbound	ECL and interest postings at month-end + events	Tie-out to warehouse

Every adapter implements: idempotency key, timeout + retry with jitter, circuit breaker, dead-letter queue, structured error taxonomy (TIMEOUT, REJECTED, SUSPECT_FRAUD, PARSE_FAIL), and a contract test against a recorded fixture. Lower environments never call production bureaus with real customers.

16.8 Decision engine internals

The decide() call is an orchestration, not a single model. Order: load application snapshot → policy knockouts (rules) → require gates G1–G5 → fetch or compute features → application PD + collection propensity + fraud already on file → affordability → limit → price → compose CREDIT_DECISION_RECORD → persist + emit. Rules live in a versioned policy pack (JSON/YAML or a BRE) promoted like code. Models are invoked only through model-platform so that model_id and feature_snapshot_id are always stamped.

Synchronous latency budget (p95, origination decide path excluding vendor): < 400 ms in-process once bureau and KYC results are already on the file. If a vendor is still running, the BFF waits on a job, not on an open HTTP call stacked five vendors deep.

16.9 Batch and near-real-time jobs

Job	Trigger	Input	Output
Day-1 ECL	AccountBooked	origination PD/LGD/EAD	ifrs9_position Stage 1
EWS score all performing	Daily 01:00 + event on bounce	payments, AA refresh, bureau delta	ews_score + band events
Behavioural PD refresh	Monthly / event	MOB performance features	credit_score type=BEHAVIOURAL
SICR + official ECL	Month-end close calendar	book snapshot + scenarios	ecl_run, GL extract
Intra-month ECL flash	Weekly	same engine, unofficial	risk pack
Sanctions rescreen	Nightly + list update	active parties	new hits → workflow

Job	Trigger	Input	Output
Re-KYC due	Nightly	refresh SLA	tasks
Collection queue build	Hourly / after score	cases + propensity	dialler / agent lists
Warehouse load	Continuous CDC + nightly snapshot	OLTP + events	analytics marts
Model monitoring	Daily	scores vs outcomes lagged	PSI, Gini, alerts

16.10 Security architecture

- Identity: OIDC for staff; customer auth separate (OTP + device). Staff roles: onboarding analyst, underwriter, override approver, collector, SIU, finance closer, model validator, platform admin. Least privilege via RBAC + resource attributes (branch, product).
- Service-to-service: mTLS or mesh identity. No shared DB users across services.
- Data classes: PUBLIC, INTERNAL, CONFIDENTIAL, RESTRICTED (PII, biometrics, bureau). RESTRICTED columns encrypted; access logged to audit-ledger.
- National IDs and raw bureau files: tokenise; analytics uses party_key. Documents scanned for malware on ingest; content-type allowlist.
- Secrets rotation 30–90 days; vendor keys per environment. Break-glass documented.
- Network: private subnets for data stores; egress allowlist to vendors; no SSH bastion without PAM recording.
- Change: policy parameters and model thresholds change via signed promotion, not by updating a row in production UI without dual control.
- Residency: keep PII and documents in-region. Warehouse in same region.

16.11 Environments and promotion

Environment	Data	Vendors	Purpose
dev	Synthetic fixtures	Simulators only	Engineers
test / CI	Synthetic + contract fixtures	Simulators + recorded HTTP	PR checks
uat / preprod	Anonymised production-shaped	Vendor UAT creds	Credit/ops acceptance
prod	Live	Live	Business
prod-analytics	Masked replica / warehouse	None	Data science; no path back to customers

Promote: Git → CI (unit, contract, schema compat) → build image → deploy to test → migrate → UAT sign-off for policy/model → prod canary → full. Database migrations expand-then-contract. Event schemas are additive; breaking changes get a new event_type or schema_version with dual publish.

16.12 Observability and operations

- Every request carries correlation_id from gateway; every event carries it. Trace origination as one distributed trace across BFF, KYC, fraud, decision.
- RED metrics per service (rate, errors, duration) plus business SLIs: KYC p95, decide p95, book success, EWS freshness, ECL job success, payment event lag.
- Alert on: vendor error budget burn, outbox lag, consumer lag, EWS job missed SLA, ECL run fail before close, poison messages in DLQ, PSI breach, disk on object store.
- Runbooks: bureau down (fail closed vs cached soft-pull policy), AA consent storm, CBS book timeout (compensate, do not double-disburse), Kafka partition hot key (party_id hash).
- Reconciliation: daily CBS balance vs account table vs warehouse; ECL stock vs GL; document hash inventory vs object store.
- Backups: PITR on OLTP; object-store versioning; tested restore quarterly. Event log is not the backup of OLTP.
- Capacity: score batch is CPU/RAM heavy — isolate EWS/ECL workers from origination pods so month-end does not kill take-up.

16.13 API conventions

- External (partner/channel): REST, JSON, idempotency-key header on POST, problem+json errors, cursor pagination, ISO-8601 timestamps in UTC.
- Internal: REST or gRPC; same idempotency for money and vendor calls.
- Version /v1; additive fields only. Deprecate with a sunset header.
- AuthZ checked in the service, not only at the gateway.
- Example resources: /v1/applications, /v1/applications/{id}/decision, /v1/accounts/{id}/ews, /v1/accounts/{id}/ecl, /v1/collection-cases.

16.14 Reference technology choices (swappable)

Concern	Sensible default	Avoid
Language (domain)	Java/Kotlin or Go for SoR; Python for ML workers	Five languages on the customer path
OLTP	PostgreSQL 16+	Mongo as ledger of balances
Bus	Kafka (or cloud equivalent)	DB polling as the only integration
Objects	S3-compatible + KMS	App server local disk
Orchestration of long KYC	Temporal / Camunda / durable workflow	Sleeping HTTP threads
Model serve	ONNX/sklearn in model-platform; not Jupyter on prod	Credit officer pastes score from Excel
IaC	Terraform/Pulumi + GitOps	Clickops in the cloud console
Containers	Kubernetes with HPA on origination; Jobs for ECL	ECL inside the API replica set

16.15 Deployment sketch

Per environment: EKS/GKE/AKS or equivalent. Namespaces: edge, origination, risk, servicing-adapter, data, jobs. Separate node pool for scoring/ECL. Ingress only to gateway. Managed

Postgres multi-AZ. Kafka multi-AZ. Object store private. CI runners have no prod credentials; deploy role is assumed by GitOps controller. Platform admin access via PAM, not standing kubeconfig on laptops.

16.16 What the system admin actually runs day-to-day

- Watch golden signals and business SLIs, not only CPU.
- Vendor status page + our adapter error rates.
- Outbox and consumer lag dashboards.
- Certificate and secret expiry calendar.
- Month-end: freeze window for ECL close; confirm job completed and GL file hash handed to finance.
- After model or policy promote: confirm decide() error rate and distribution of grades did not cliff.
- Quarterly: restore test, access review, fire-drill bureau-down and CBS-timeout.

16.17 Suggested repo / ownership map

Repo or pack	Owner	Contains
los-origination	Lending engineering	application, BFF, agreement
los-party-kyc	Identity engineering	party, screening adapters
los-fraud-docs	Risk engineering	documents, forensics, fraud
los-decision	Credit engineering	rules, decide API, pricing
los-servicing-adapter	Core / payments	CBS anti-corruption layer
los-ifrs9	Finance engineering + risk	ECL engine, staging
los-ews-collections	Collections engineering	EWS, cases, treatments
los-model-platform	ML platform	registry, serving, features
los-events-contracts	Platform	Avro/JSON schemas, compatibility CI
los-infra	SRE	Terraform, Kafka topics, alerts
los-policy-packs	Credit risk (with eng review)	Thresholds, grids, knockouts

17. Closing note

A loan system is not a form plus a bureau score. It is a chain of evidence (who is this?), integrity (is the file real?), capacity and willingness (will they pay?), price and capital (are we paid for the risk?), continuous observation (has the risk moved?), and intervention (can we still collect, and at what cost?). The schema above is the minimum object model, metric set, and runtime architecture that lets that chain be audited, improved, and operated.

Engineering should implement the service catalog, event envelope, outbox, and adapters first. Credit and finance should own policy packs and model versions that deploy through the same pipeline. If those two planes stay in separate languages, the architecture will drift into another siloed scoring spreadsheet within a year.